Pin Map + Wiring Notes – Spiral Lamp

1. Pin Map (What Goes Where)

LED Strip

Function	Arduino Pin	Notes
LED data (DIN)	**D9**	WS2812 / NeoPixel strip, 60 LEDs, GRB @ 800 kHz
LED 5 V	—	From external 5 V supply (not from Arduino 5 V if you can avoid it)
LED GND	—	Tie to Arduino GND and power supply GND

Buttons & Switches

All of these are simple **digital inputs**. Use momentary pushbuttons for the arcade buttons and latching switches for the rest.

Control / Label	Function in Code	Arduino Pin
Main Rocker	`PIN_ROCKER` – base on/off & Jumbo fade	**D7**
Modifier Toggle	`PIN_TOGGLE` – changes Jumbo / Wave / Rainbow behavior	**D6**
Red Arcade Button	`PIN_LEFT` – delete dots / Wave reverse-hold / Lava bottom color cycle	**D8**
White Arcade Button	`PIN_RIGHT` – spawn dots / Wave pause / Lava top color cycle	**D4**
Rainbow Key Switch	`PIN_KEY` – enables Rainbow mode	**D5**
Wave Cover Switch	`PIN_WAVE` – enables Wave cover animation	**D10**
Chaos / Lava Switch	`PIN_CHAOS` – enables full Lava Lamp takeover	**D11**

Knobs / Potentiometers

Control	Function in Code	Arduino Pin
Brightness Fader	`PIN_FADER` – global brightness (0–100%)	**A0**
Speed Knob	`PIN_SPEED_KNOB` – timing/speed control (Rainbow, Wave, Dots, Lava)	**A1**

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2. Basic Wiring Pattern

LED Strip

- **5 V** from your external supply → LED strip +5 V pad.
- **GND** from the same supply → LED strip GND **and** Arduino GND.
- **DIN** → Arduino **D9** through a small series resistor (≈330–470 Ω).
- Put a big electrolytic cap (≈1000 µF, 6.3 V or higher) across 5 V and GND near the strip.

Buttons & Switches (Digital Inputs)

General pattern:

- One side of the switch/button → **Arduino pin**.
- Other side → **GND**.

The sketch turns on the internal pull-ups, so the input reads:

- **HIGH** when the switch is **open** (not pressed).
- **LOW** when the switch/button is **closed** (pressed / ON).

Wire **all** of these this way:

- D7 – main rocker
- D6 – modifier toggle
- D4 – white button
- D5 – rainbow key
- D8 – red button
- D10 – wave switch

Chaos / Lava switch (D11):

- Wire as a 2-position switch that clearly gives **two states** to D11 (HIGH vs LOW).
- One simple approach:
  - Common → **D11**
  - Throw 1 → **5 V** (Chaos ON)
  - Throw 2 → **GND** (Chaos OFF)

Pots / Fader (Analog Inputs)

For both the brightness fader (A0) and speed knob (A1), use standard 3-pin wiring:

- One outer leg → **5 V**
- Other outer leg → **GND**

- Center wiper → **A0** (fader) or **A1** (speed knob)

10 kΩ linear pots work great.

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3. Quick “What Goes Where” Checklist

If you’re staring at your Arduino and strip:

1. **LED DIN** → D9
2. **Main rocker** → between D7 and GND
3. **Modifier toggle** → between D6 and GND
4. **White button** → between D4 and GND
5. **Red button** → between D8 and GND
6. **Rainbow key switch** → between D5 and GND
7. **Wave switch** → between D10 and GND
8. **Chaos / Lava switch** → D11 to either 5 V (Chaos ON) or GND (Chaos OFF)
9. **Brightness fader** → wiper to A0, ends to 5 V and GND
10. **Speed knob** → wiper to A1, ends to 5 V and GND
11. **All grounds tied together**: LED GND, Arduino GND, power supply GND, switch grounds.

That’s the wiring reference I’d keep next to the bench so you don’t have to decode it from the code or the video.